

Principles and Applications of Database 数据库原理与应用

Experiment: SQL Queries on the Movie-Rating Database

Due Date: June 26, 2026

Objective

The objective of this experiment is to develop practical skills in writing and executing SQL queries involving selection, projection, joins, aggregation, grouping, sorting, subqueries, and handling of NULL values.

Database Schema and Data

The Movie-Rating database schema and sample data are attached to this experiment sheet. Students must use the attached schema and data to create and populate the database before attempting the questions.

The database consists of the following relations:

- **Movie** (mID, title, year, director)
- **Reviewer** (rID, name)
- **Rating** (rID, mID, stars, ratingDate)

Important: Students must use the attached dataset exactly as provided. Do not modify the data or use a different dataset.

Requirements

1. Create and populate the database using the attached schema and sample data.
2. Write SQL queries to answer Questions Q1–Q10.
3. Execute each query using SQLite or another approved SQL database system.
4. For each question, provide:
 - The SQL query.
 - The query result/output as a screenshot in light theme.
5. Compile all questions into a report.
6. Print the report and submit a hard copy on or before 11:00 am on June 26, 2026 at 东 6E313.
7. Reports must be typed and professionally formatted.

Submission Format

For each question, organize your report as follows:

1. Question Number
2. SQL Query
3. Query Result (Screenshot)

Questions

Q1. Find the titles of all movies directed by Steven Spielberg.

Q2. Find all years that have a movie that received a rating of 4 or 5 stars. Sort the years in increasing order.

Q3. Find the titles of all movies that have no ratings.

Q4. Some reviewers did not provide a date with their rating. Find the names of all reviewers who have ratings with a NULL value for the date.

Q5. Return the ratings data in a more readable format showing:

- Reviewer name
- Movie title
- Number of stars
- Rating date

Sort the results by reviewer name, movie title, and number of stars.

Q6. For all cases where the same reviewer rated the same movie twice and gave it a higher rating the second time, return the reviewer's name and the title of the movie.

Q7. For each movie that has at least one rating, find the highest number of stars that movie received.

Return:

- Movie title
- Highest rating received

Sort the results by movie title.

Q8. For each movie, return the title and the “rating spread”, that is, the difference between the highest and lowest ratings given to that movie. Sort by rating spread from highest to lowest, then by movie title.

Q9. Find the difference between the average rating of movies released before 1980 and the average rating of movies released after 1980.

Important:

- First calculate the average rating for each movie.
- Then calculate the average of those movie averages for movies released before 1980 and after 1980.
- Do not calculate a single overall average rating before and after 1980.

Q10. Find the names of all reviewers who have rated three or more different movies.

Return:

- Reviewer name
- Number of movies reviewed

Sort the results by the number of movies reviewed in descending order.

Marking Scheme

Item	Weight
Correct SQL Queries	50%
Correct Query Results	50%
Total	100%

Note: Students must execute all queries and include the corresponding query results in their report. Reports that contain only SQL statements without results will not receive full marks.

End of Experiment